

Api 650 Welded Storage Tanks

Page 1: Storage Tank Design and Shell Calculations

API 650 is the global standard for welded steel tanks for oil storage. It governs vertical, cylindrical, aboveground, closed- and open-top tanks of various sizes and capacities for operating pressures up to 2.5 psi (17 kPa) and temperatures up to 93 deg C.

The tank shell thickness is designed block-by-block using the One-Foot Method: $t = (4.9 * D * (H - 0.3) * G) / Sd + CA$, where t is the shell thickness in mm, D is the tank diameter in meters, H is the design liquid height in meters, G is the specific gravity of the liquid, Sd is the allowable design stress of the steel plate, and CA is the corrosion allowance.